

6.2 Nitrogen compounds, polymers and synthesis

This section focuses on organic nitrogen compounds, including amines, amides and amino acids. Chirality and optical isomerism is also introduced.

Condensation polymerisation is also introduced and compared with addition polymerisation.

The importance of carbon–carbon bond formation in organic synthesis is stressed. Learners are also able

to consider multi-stage synthetic routes towards an organic product.

This module allows learners many opportunities to further develop their organic practical skills, especially in preparing and purifying organic solids, including recrystallisation and determination of melting points.

6.2.1 Amines

Learning outcomes	Additional guidance
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Learners should be able to demonstrate and apply their knowledge and understanding of:

Basicity and preparation of amines

- (a) the basicity of amines in terms of proton acceptance by the nitrogen lone pair and the reactions of amines with dilute acids, e.g. HCl(aq) , to form salts

Comparison of basicity of different amines **not** required.
Restricted to inorganic acids.

(b) the preparation of:

- (i)** aliphatic amines by substitution of haloalkanes with excess ethanolic ammonia and amines
- (ii)** aromatic amines by reduction of nitroarenes using tin and concentrated hydrochloric acid.

Including formation of primary amines from ammonia and secondary/tertiary amines from amines.
See also reduction of nitriles (**see 6.2.4 c**).